Renewable-Powered Desalination: A Sustainable Alternative to Conventional Water Treatment Systems

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***Abstract*—**Renewable-powered seawater reverse osmosis (SWRO) is widely promoted for low-carbon water supply, and flexible, renewable-following operation is often reported to lower cost. However, the underlying dispatch models treat the plant as a fixed power-to-water converter and omit membrane degradation, although published experiments link intermittent operation to faster fouling. To address this gap, we couple a plant-scale biofouling and irreversible-degradation model to an hourly dispatch and storage co-optimization, driven by NASA POWER data for the Al Taweelah SWRO site (mean AOD_{550} = 0.545). We introduce the cycle tolerance N_{tol}, the number of start/stop cycles a membrane sustains before its irreversible fouling budget is exhausted, as the measurable parameter governing flexible operation. In a controlled comparison of degradation-blind and degradation-aware dispatch, blind load-following raises the renewable fraction from 39.5% to 52.3% but shortens membrane life from 2.06 yr to 0.53 yr and increases the levelized cost of water by 45% (from 0.992 USD/m³ to 1.434 USD/m³), whereas the aware controller selects zero flexibility. Flexibility becomes economic only above $N_{tol}^* \approx 4201$ cycles, equivalent to 11.7 yr of daily cycling, and its largest benefit is 198 times smaller than the penalty for pricing it blind. A steady-baseload plant saturates at a 43.4% renewable fraction, and its photovoltaic capacity requires a carbon price of 38 USD/t CO_2 to 155 USD/t CO_2 at an intermediate grid emission factor. Membrane cycle tolerance, rather than the cost of solar generation or storage, is therefore the binding constraint, and a bench experiment to measure it is specified.

***Index Terms*—**Arabian Gulf, biofouling, membrane degradation, renewable integration, SDG 6, seawater reverse osmosis, techno-economic optimization, water–energy nexus.

I. Introduction

Desalination is the foundation of water security in the Gulf. In the United Arab Emirates it supplies over 90% of drinking water, consumes approximately 15% of national electricity, and accounts for more than 22% of energy-related carbon dioxide emissions [1]. Any credible pathway to Sustainable Development Goal 6 in the region therefore runs through both the energy intensity and the energy source of desalination, placing the problem at the intersection of the UN Water pillars of Water for People, Water for Prosperity, and Investments for Water.

It is tempting to frame the challenge as reducing the specific energy consumption of reverse osmosis. That framing is largely exhausted. On the osmotic relation used in this paper (Eq. (1)), the reversible least work of separating seawater at 35 g/L and 50% recovery is 1.06 kWh/m³, and a single-stage RO train with ideal pumps and energy recovery cannot operate below 1.53 kWh/m³, the single-stage thermodynamic limit analyzed by Li [2]. A benchmarking of 64 facilities quantifies the further savings available only by combining state-of-the-art measures, including semi-batch operation [3], and a Gulf pilot projects 3.11 kWh/m³ to 3.13 kWh/m³ for a full-scale plant on high-salinity feed [4].

Energy is also a smaller share of cost than commonly assumed: in our own cost model of a conventional Gulf plant, electricity is 21% of the levelized cost of water, against 37% for capital recovery.

The live question is therefore not how little energy RO can use, but where that energy comes from and when it is consumed. Co-locating photovoltaics with SWRO is already commercial practice: Al Taweelah

in Abu Dhabi, the world's largest SWRO facility at 909,200 m³/d, is partly powered by a co-located solar park of about 69 MW_p [5], with solar expected to supply at least 30% of the project's electricity capacity within eight years and a target of 55% [6]. Measured in energy rather than capacity, the same array supplies only about 9% of the plant's annual electricity at this site's yield on our energy balance. The gap between capacity share and energy share is the daylight-absorption limit this paper examines: we ask what prevents the energy share from rising, and what would move it.

A. Why the existing answer is insufficient

Two bodies of work speak to this question, and they do not speak to each other.

The first is renewable dispatch optimization, in which the desalination plant is modeled as a flexible load or hybrid generator–load resource. Variable-power operation has been developed to match RO load to intermittent solar supply [7], [8]; desalination flexibility has been offered into demand-response and frequency-regulation markets [9]; an adaptive robust optimization of an RO plant's demand-response intervals reports cost reductions of over 20% [10]; and a broader framework values energy flexibility from water systems generally [11]. The state of the art in this line, Alahmed et al. [12], derives threshold-based optimal dispatch policies for renewable-integrated desalination. Critically, that work states its own limitations: energy storage is not represented, and the water–electricity conversion functions are "rather simplified", because in practice "energy-use depends on ambient conditions such as salinity and temperature". Membrane fouling and irreversible degradation do not appear in the model at all, and no Gulf-specific salinity or temperature is used.

The second is the membrane fouling literature, and its evidence on intermittent operation is not reassuring. Fouling degrades flux, raises feed pressure and energy use, and shortens membrane life [13], [14]. Laboratory work on brackish groundwater finds that intermittent operation accelerates scaling and fouling: with antiscalant and a rinse before shutdown, permeability stays above 70% of its initial value after seven days, but without rinsing it falls below 50%, with similar behavior at pilot scale [15], [16]. A full-scale brackish plant operated intermittently for 14 years lost about half of its water permeability while its specific energy rose from 1.82 to 2.21 kWh/m³ [17], and a seawater membrane under photovoltaic-emulated variable operation lost 37% of its permeability within three days without rinsing [18]. Design-stage work has already shown that ignoring this fouling under-sizes PV-powered RO systems [19].

The evidence is not uniform: real-time imaging of gypsum scaling on synthetic brackish feed found no added flux decline from daily shutdowns [20], although that feed carried no biological load. Similarly, a 100 m³/d wind-powered SWRO plant operated in 1 h on and 1 h off cycles for more than four months (1,568 start–stop cycles) showed only slight fouling and no significant membrane damage [21], although each shutdown lasted only one hour, far shorter than the overnight stops of solar-following operation. Yet none of these results has been carried into the dispatch models above; none isolates the irreversible loss per start–stop cycle over hundreds of cycles against a continuous control; and none concerns warm, hypersaline, biologically active Gulf feedwater.

The Arabian Gulf is the harshest case. It is a shallow, semi-enclosed sea with an average depth of 35 m and average salinity of 39 psu, whose surface temperature reaches 36 °C in summer and falls below 15 °C in winter [22]. A Gulf pilot plant treating feed of 47.5 g/L to 53.5 g/L, with turbidity from algal blooms, concludes that current SWRO technology is not energy-efficient for these conditions [4], and an operating Gulf SWRO plant has documented serious membrane biofouling [23]. The degradation penalty of intermittent operation is therefore likely to be larger here than in the brackish studies above, and its economic consequence for dispatch has not been quantified.

B. Approach and contributions

We couple the two literatures. We construct a plant-scale model in which (i) RO specific energy depends on feed salinity, temperature, and load fraction through explicit osmotic and permeability relations; (ii) fouling accumulates as reversible and irreversible hydraulic resistance whose growth responds to temperature via a Q_{10} law, to feed biological load, to crossflow shear at part load, to stagnation during idle hours, and to discrete start/stop transients; and (iii) the resulting degradation feeds back into both specific energy and membrane replacement interval. This is embedded in an hourly dispatch-and-storage co-optimization over a 25 yr economic horizon, driven by measured irradiance, air temperature, and aerosol optical depth.

Our contributions are:

1. A coupled dispatch–degradation model for Gulf SWRO (Section III), benchmarked against a published full-scale Gulf plant projection and checked against standard engineering values (Table I).
2. The cycle tolerance N_{tol} as the governing parameter (Section III-H). The economics of flexible renewable-following operation are controlled by a single quantity with a direct bench-scale measurement protocol; reporting the frontier in those units makes the conclusion falsifiable.
3. A controlled quantification of the cost of ignoring degradation (Section IV-B), using dispatch policies that share structure and foresight and differ only in one objective weight.
4. A demonstration that the result is structural, not marginal (Section IV-C). Flexibility is dominated across the whole plausible range of cycle tolerance, across biofouling severity, and across tariff spread; removing restart damage entirely is not sufficient to reverse it.
5. Identification of membrane cycle tolerance as the binding constraint on renewable penetration (Section IV-F), rather than the cost of solar generation or storage.
6. An explanation, rather than a criticism, of observed operator behavior (Section V), together with the implicit carbon price that justifies existing PV deployment.
7. A specified experiment (Section VII) to measure the decisive unmeasured quantities.

We state at the outset what this study is not. It is not a validated plant model. It is benchmarked against published values and standard engineering checks, not against proprietary operating records, and 34 of its 52 parameters are explicit assumptions. We therefore report conclusions as functions of those assumptions rather than as point predictions, and Section VI states the consequences plainly.

II. Related Work

Energy limits of reverse osmosis. The thermodynamic framing of SWRO energy use is settled [2], [3]. Process intensification through batch and semi-batch configurations offers substantial gains (up to 37% for closed-circuit RO and 64% for batch RO relative to continuous operation [24]), but those figures are obtained at high recovery on brackish feed, and the corresponding headroom at seawater salinity is considerably smaller. We therefore treat single-stage continuous SWRO with isobaric energy recovery as the relevant baseline technology and make no claim of process-level energy reduction.

Renewable integration and techno-economics. Solar-driven SWRO in the Gulf region has been assessed repeatedly, including location-resolved photovoltaic assessments for Saudi Arabia [25] and comparisons of battery, hydrogen, and pumped-hydro storage for renewable-powered RO [26], [27]; broader reviews survey the configuration space [28]. Closest to our setting, Janowitz et al. [29] evaluate large-scale SWRO operated to follow a photovoltaic power profile in Jordan, finding a water cost of about 2.5 USD m^{-3} at 46% water production availability against 1.8 USD m^{-3} for full-load operation at the same 46% photovoltaic power share, with safe operating windows further restricted by seasonal temperature and salinity. They reach a conclusion consistent with ours through capacity utilization alone,

without modeling membrane degradation; adding the degradation mechanism studied here would be expected to widen that gap. This literature is mature, and a further static techno-economic comparison would add little. Our contribution is orthogonal: we hold the configuration space fixed and change what the objective function knows about the membrane.

Flexible operation and demand response. Desalination plants are increasingly treated as grid-flexible assets [9]–[11], with Alahmed et al. [12] providing analytical dispatch policies for renewable-integrated plants. Across this literature the plant is a static converter and degradation is out of scope, typically acknowledged as future work. Our model is a direct response to that stated gap.

Fouling, intermittency, and cost. The performance and cost consequences of fouling are reviewed extensively [13], [30], and the resistance-in-series framework of Ling et al. [31] provides the structural basis we adopt, though their analysis is at CFD scale and does not resolve temperature or crossflow dependence. Most experimental studies of intermittent operation find a penalty that rinsing and antiscalant mitigate but do not remove [15], [16], [18], although an imaging study of gypsum scaling found no shutdown penalty [20], and a wind-powered seawater plant showed no significant membrane damage after 1,568 short start–stop cycles [21]; a full-scale intermittent plant shows substantial long-term permeability decline [17]; and design-stage coupling of fouling to PV-RO sizing establishes that neglecting fouling under-sizes systems [19]. To our knowledge no study couples Gulf-parameterized biofouling dynamics to dispatch-level renewable co-optimization, which is the space this paper occupies.

III. Methods

A. Notation

S denotes salinity (g/L), subscripts f feed and c concentrate; T temperature (°C); r recovery ratio; π osmotic pressure (bar); P_f feed pressure (bar); A water permeability (L/(m² h bar)); J permeate flux (L/(m² h)); R_{rev}, R_{irr} reversible and irreversible fouling resistance (m² h bar/L); L load fraction of RO nameplate; B dimensionless feed biofouling index (B = 1 open ocean); λ shadow price of irreversible fouling; N_{tol} cycle tolerance; SEC specific energy consumption (kWh/m³); LCOW levelized cost of water (USD/m³).

B. Seawater and membrane relations

Osmotic pressure is a van't Hoff expression for a NaCl-equivalent electrolyte with osmotic coefficient φ,

$$\pi = \varphi\, i\, \frac{S}{M_{\mathrm{NaCl}}}\, R\, T_K \tag{1}$$

with i = 2, M_{NaCl} = 58.44 g/mol, R = 0.083145 L bar/(mol K). We set φ = 0.93, giving 27.61 bar at 35 g/L and 25 °C, against about 29.7 bar for φ = 1. This is a standard engineering value rather than a calibration; comprehensive seawater property correlations [32] would replace Eq. (1) in a refined model. Concentrate salinity follows from mass balance, $S_c = S_f(1 - r(1 - R))/(1 - r)$, with rejection R = 0.995.

Permeability is temperature-corrected by the standard manufacturer normalization

$$\mathrm{TCF}(T) = \exp\left\{K_e \left(\frac{1}{T_0} - \frac{1}{T_K}\right)\right\} \tag{2}$$

with K_e = 2700 for polyamide seawater membranes (reported range 2500–5500). Equation (2) gives 0.857 at 20 °C and 1.342 at 35 °C: warm water permeates substantially more easily, a point that proves consequential in Section IV-A.

Fouling enters as additive hydraulic resistance in the resistance-in-series form of Ling et al. [31],

$$\frac{1}{A_{\mathrm{eff}}} = \frac{1}{A_{\mathrm{clean}}(T)} + R_{\mathrm{rev}} + R_{\mathrm{irr}} \tag{3}$$

C. Site and resource data

The reference site is Al Taweelah, Abu Dhabi (24.750° N, 54.700° E). Hourly global horizontal and clear-sky irradiance, air temperature, wind speed and relative humidity, and daily aerosol optical depth at 550 nm, were retrieved for calendar year 2023 from the NASA POWER reanalysis [33]. The site receives 2009 kWh/(m² yr) of global horizontal irradiance with air temperature spanning 15.7 °C to 43.9 °C. Mean AOD_{550} is 0.545, with 95th percentile 0.99 and maximum 1.38. This aerosol load is roughly five times that of a low-aerosol site and is the basis for the soiling model below.

Feed pressure to sustain design flux is

$$P_f = \pi(S_c, T) + \frac{J}{A_{\mathrm{eff}}} + \Delta P_{\mathrm{fric}} \tag{4}$$

evaluated at the concentrate condition, the conservative engineering convention since the last element must still produce.

Gulf seawater temperature at the intake is represented as a sinusoidal seasonal cycle between 20 °C and 35 °C. This is narrower than the surface range reported for the basin, which reaches 36 °C in summer and falls below 15 °C in winter [22]; the consequence is examined in Section VI. Design feed salinity is 45 g/L, swept over 39 g/L to 47 g/L, between the basin average of 39 psu [22] and the 47.5 g/L to 53.5 g/L recorded at a Gulf pilot intake [4]. To keep multi-year hourly simulation tractable we model one representative train serving a fixed water obligation of 41,823 m³/d, one twentieth of the plant at 92% mean utilization.

D. Specific energy consumption

With isobaric energy recovery on the brine stream, energy per cubic meter of permeate is

$$\mathrm{SEC}_{\mathrm{proc}} = \kappa \left[\frac{P_f}{r\,\eta_{\mathrm{HPP}}} - \eta_{\mathrm{ERD}}\, P_b\, \frac{1-r}{r} \right] \tag{5}$$

where $P_b = P_f - \Delta P_{\mathrm{fric}}$, $\kappa = 0.0277778$ kWh/(bar m³), $\eta_{HPP} = 0.85$ (an assumption; Karameldin et al. [34] use 0.80), and $\eta_{ERD} = 0.94$, within the 92–96% net transfer efficiency reported for positive-displacement energy recovery devices [34]. Part-load pump efficiency is derated quadratically, $\eta_{HPP}(L) = \eta_{HPP}[1 - 0.15(1 - L)^2]$, so deep load-following carries an efficiency penalty independent of any fouling effect. Plant-total SEC adds 0.50 kWh/m³ for intake, pre-treatment, post-treatment, and product pumping.

E. Photovoltaic generation with aerosol soiling

PV output uses a NOCT cell-temperature estimate with power-temperature coefficient −0.34 %/K and inverter clipping at DC/AC ratio 1.2. Soiling accumulates between washes in proportion to aerosol optical depth,

$$\ell(d) = \min\left(\ell_{\max},\ k_{\mathrm{soil}} \sum_{j=d_0}^{d} \mathrm{AOD}_{550}\,(j)\right) \tag{6}$$

resetting at each wash, with $\ell_{max} = 0.25$, a 14 d wash interval, and 0.35 USD/m² per session [1]. This makes the Gulf dust burden an explicit cost rather than a lumped derate.

F. Fouling and degradation dynamics

This is the component absent from the dispatch literature. Resistance accrues during operation at

$$\dot{R}_{\mathrm{op}} = k_{\mathrm{bio}}\, Q_{10}^{(T-T_{\mathrm{ref}})/10}\, B\, L^{-\alpha} + k_{\mathrm{scale}}\, \frac{r}{1-r} \tag{7}$$

and during idle hours at

$$\dot{R}_{\mathrm{idle}} = k_{\mathrm{bio}}\, Q_{10}^{(T-T_{\mathrm{ref}})/10}\, B\, m_{\mathrm{idle}} \tag{8}$$

Three mechanisms here are specific to this study. The exponent α captures weakening shear removal of nascent biofilm as crossflow velocity falls at part load. The multiplier m_{idle} captures stagnation, during which there is no crossflow at all. Each start applies a discrete increment δR representing the flux and pressure transient that compacts deposits. A fraction f_{irr} of continuously-accrued resistance and a larger fraction f_{irr}^{start} of restart-induced resistance are irreversible.

Clean-in-place is triggered when normalized permeability falls below 0.85 and at least 720 h have elapsed since the last cleaning. The minimum-interval condition matters: a pure threshold rule thrashes as irreversible resistance accumulates, producing physically absurd cleaning frequencies. Replacement occurs when cleaning can no longer restore normalized permeability to 0.70, or at a 5 yr ceiling.

We set k_{bio} so that with B = 1 at 25 °C the model reaches the cleaning trigger in 4.7 months; this is an assumption, not a calibration to plant data. Reported intervals span a wide range: a full-scale SWRO plant reports a 20-day cleaning frequency alongside a 40% decline in normalized flux over seven years [35], while a Gulf pilot with a two-stage high-recovery design ran for more than four months without cleaning [36]. Faster fouling is spanned by the biofouling-index sweep of Fig. 4; at B = 2 and 30 °C the model predicts a cleaning interval of approximately 1.6 months. Idle fouling runs 2.26 to 2.44 times the full-load rate across the temperature range, and part-load operation at L = 0.4 runs 2.03 times the full-load rate.

G. The shadow price of degradation

To price degradation inside the dispatch decision, note that replacement is triggered at the irreversible resistance R* at which normalized permeability equals p_{rep},

$$R^* = \frac{1}{A_{\text{clean}}}\left(\frac{1}{p_{\text{rep}}} - 1\right) \tag{9}$$

Spreading membrane set cost C_{mem} over that budget yields

$$\lambda = \frac{C_{\text{mem}}}{R^*} \tag{10}$$

the expected cost of each unit of irreversible fouling. Equation (10) is what converts membrane life from an engineering afterthought into a term in the objective function.

H. Cycle tolerance

The per-restart increment δR is a model coefficient with no direct physical interpretation, and it is unmeasured for Gulf seawater. We therefore recast it as the *cycle tolerance*

$$N_{\text{tol}} = \frac{R^*}{f_{\text{irr}}^{\text{start}}\,\delta R} \tag{11}$$

the number of start/stop cycles a membrane sustains before its irreversible budget is exhausted. This has three advantages. It is dimensionless and interpretable: solar-following implies approximately 360 starts per year, so $N_{tol}/360$ is directly the membrane life under daily cycling. It is *measurable*: cycle a seawater element daily on Gulf feed and record the permeability not recovered by cleaning. And reporting the frontier in these units makes the central claim falsifiable by a single bench experiment.

I. Dispatch policies and the controlled comparison

Each hour the controller selects a load fraction $L \in \{0\} \cup [L_{min}, 1]$ with $L_{min} = 0.4$, subject to meeting the water obligation from the product tank. Energy is sourced from PV first, then battery, then grid.

A valid test of whether pricing degradation changes the answer requires policies differing *only* in that respect. We therefore use one dispatch structure with two knobs. The controller minimizes

$$\min_{L} \left[\underbrace{c_e(L)}_{\text{energy}} + w\,\lambda\,\underbrace{\Delta R_{\mathrm{irr}}(L)}_{\text{degradation}} - \underbrace{v_w(L)}_{\text{water value}} \right] \tag{12}$$

where w = 0 gives the degradation-blind controller assumed by the dispatch literature and w = 1 the degradation-aware controller proposed here. Both receive the same 24 h-ahead PV signal, and both search the same flexibility-aggression parameter, which *includes zero*. Including the degenerate no-flexibility case is essential: if steady operation is economically correct, the policy family must be able to express it, otherwise the experiment measures the heuristic rather than the physics. Both controllers are evaluated on *true* LCOW. The blind designer optimizes a cost function from which membrane replacement is absent and then pays the real bill, which is precisely the error we quantify.

TABLE I

Benchmarks and consistency checks. Rows marked ✓ compare against a verified published value; the others are consistency checks against standard engineering values we have not individually sourced. Gulf-condition outputs are model predictions, not calibration targets.

Quantity	Model	Reference
Process SEC, Gulf summer (kWh m^{-3})	3.17	✓ 3.11–3.13 projected, full-scale Gulf plant [4]
Process SEC, ocean (kWh m^{-3})	2.71	✓ above 1.794, lowest plant value on record [37]
Feed pressure, Gulf (bar)	72.7	✓ above ≈65 for standard designs, below ≈83 element rating [34]
ERD net efficiency	0.94	✓ 0.92–0.96, positive-displacement devices [34]
π at 35 g/L, 25 °C (bar)	27.61	consistency check, φ = 0.93
Cleaning interval, ocean, B=1 (months)	4.7	not calibrated; 20 days at a full-scale plant [35], >4 months at a Gulf pilot [36]

J. Economics

Levelized cost of water is

$$\mathrm{LCOW} = \frac{\mathrm{CRF}\cdot\mathrm{CAPEX} + \mathrm{OPEX}_{\mathrm{ann}}}{V_{\mathrm{delivered}}}, \qquad \mathrm{CRF} = \frac{i\,(1+i)^n}{(1+i)^n - 1} \tag{13}$$

with i = 5.5% real and n = 25 yr, as used in a recent techno-economic assessment of solar SWRO at Jubail on the Gulf coast [38]. Annual operating cost comprises grid electricity net of export revenue, membrane replacement, cleaning, PV washing, and fixed operations. LCOW is charged against water actually delivered, so designs failing the obligation are penalized rather than flattered by producing less. RO over-size (nameplate divided by mean obligation) is a design variable, because spare capacity is precisely what buys the option to concentrate production into sunlit hours. Conventional plants sit near 1.09. All parameters and provenance are in Appendix A.

IV. Results

A. The Gulf seasonal tension

The model's first output is counterintuitive. Plant-total SEC is lowest in summer, at 3.672 kWh/m³ at 35.0 °C (day 225), and highest in winter, at 3.778 kWh/m³ at 20.0 °C (day 43). The temperature term of Eq. (2) dominates the osmotic term of Eq. (1): cold water permeates poorly, and that penalty outweighs its lower osmotic pressure. On this model the Gulf's heat is energetically helpful. A survey of more than 70 large-scale SWRO datasets finds the effect of temperature on plant energy use not entirely clear in practice [39]; the pattern here follows from the temperature normalization of Eq. (2) and should be read as a prediction.

Over the same cycle the biofouling rate varies by a factor of 2.65, from 7.03×10^{-5} to 1.87×10^{-4} in model units, rising with temperature through the Q_{10} law. The season with the cheapest energy is therefore the season with the fastest fouling (Fig. 1). This opposition is the mechanism underlying everything that follows: any strategy concentrating production into hot, sunny periods simultaneously concentrates it into the highest-fouling regime.

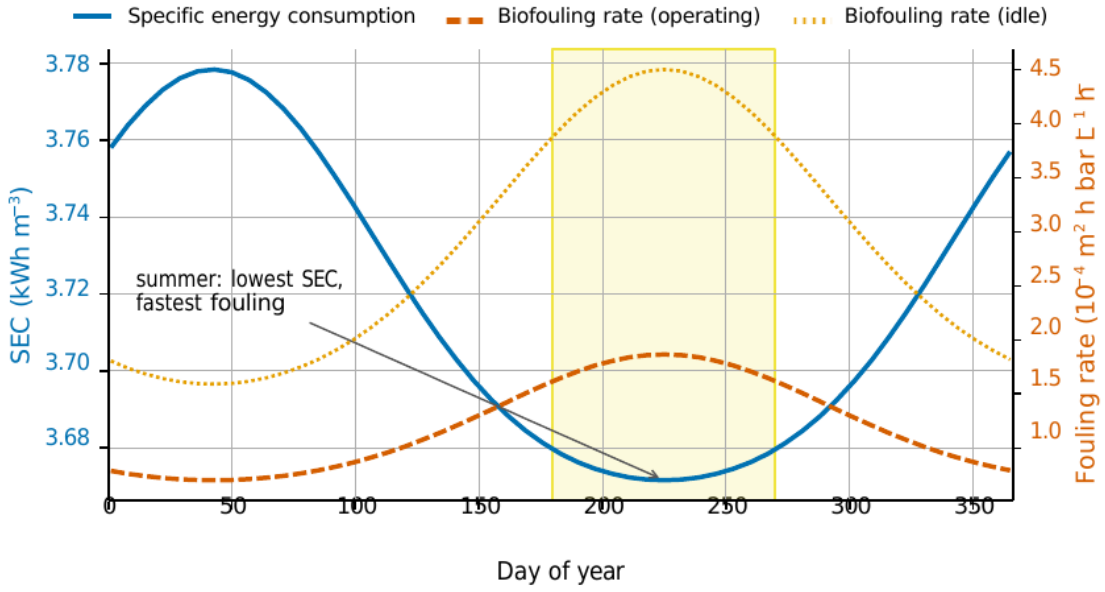


Fig. 1. Seasonal opposition of energy cost and fouling rate in the Arabian Gulf. Specific energy consumption is minimized in summer because membrane permeability rises with temperature, while biofouling rate is maximized in the same period.

B. The cost of degradation-blind dispatch

Table II reports the controlled comparison at a flexibility-enabled design (over-size 1.60, 12 h product storage, 21 MW_p PV). The degradation-blind controller behaves exactly as the dispatch literature anticipates on the energy axis: it raises renewable fraction from 39.5% to 52.3%, a gain of 12.9 percentage points, and cuts the electricity component from 0.096 USD/m³ to 0.076 USD/m³. It does so by cycling the plant daily (361 restarts per year), which collapses membrane life from 2.06 yr to 0.53 yr. The membrane cost component rises almost fourfold, from 0.158 USD/m³ to 0.610 USD/m³, and LCOW rises 45% from 0.992 USD/m³ to 1.434 USD/m³. The 0.021 USD/m³ of energy saving is overwhelmed by 0.452 USD/m³ of additional membrane cost, a ratio of 22 to one.

TABLE II
Controlled policy comparison. The two flexible controllers share dispatch structure and day-ahead foresight and differ only in the degradation weight w of Eq. (12).

	Grid base	PV + base	Flex blind	Flex aware
LCOW (USD/m³)	0.942	0.992	**1.434**	**0.992**
Renewable fraction (%)	0.0	39.5	52.3	39.5
Membrane life (yr)	2.06	2.06	**0.53**	2.06
Restarts per year	0	0	361	0
Mean SEC (kWh/m³)	3.98	3.98	3.97	3.98
Capacity factor	1.00	1.00	0.88	1.00
Cleanings per year	11.0	11.0	11.7	11.0
Membrane (USD/m³)	0.158	0.158	**0.610**	0.158
Electricity (USD/m³)	0.159	0.096	0.076	0.097
Capital (USD/m³)	0.401	0.468	0.469	0.468
Fixed O&M (USD/m³)	0.220	0.220	0.220	0.220

Given identical structure, foresight, and design space, the degradation-aware controller selects zero flexibility: it chooses the degenerate steady-baseload member of its own policy family, reproducing the PV-baseload result exactly (0 restarts per year, LCOW 0.992 USD/m³). Under Gulf conditions at the default cycle tolerance the optimal amount of load-following is none at all (Fig. 2).

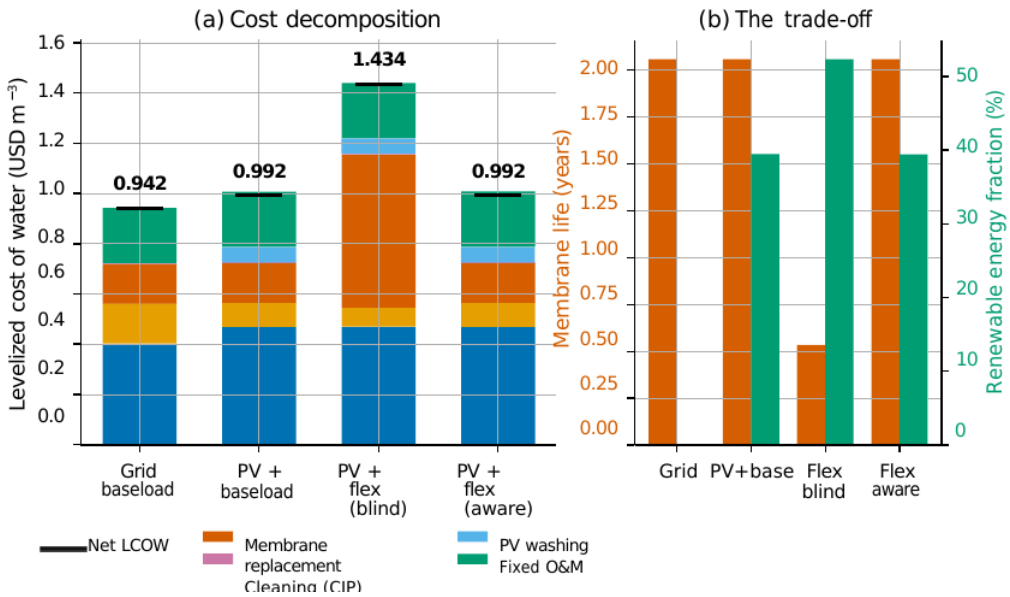


Fig. 2. (a) LCOW decomposition by policy. (b) The trade-off: renewable fraction against membrane life. Degradation-blind flexing buys renewable penetration by consuming membrane life.

C. The cycle-tolerance frontier

The preceding result depends on the per-restart degradation increment, the least-constrained parameter in the model. Section III-H recast it as cycle tolerance so the dependence can be reported directly. Table III and Fig. 3 sweep N_{tol}, re-optimizing dispatch at each value. The critical tolerance is $N_{tol}^{*} \approx 4201$ cycles, equivalent to 11.7 yr of daily cycling. This is the paper's central number, and its magnitude is the point: operating SWRO plants replace membranes after 2 yr to 5 yr [40], corresponding to 720–1800 cycles under daily start–stop. The tolerance required to make load-following economic is therefore more than twice the longest reported membrane service life. On current membrane technology the frontier is not close; it is out of reach.

Three further features of Table III matter. First, across the entire physically plausible range (N_{tol} from 250 to 4000, that is from 0.7 yr to 11.1 yr of daily cycling), the degradation-aware controller returns $\Delta = 0$ exactly. It is not trading a small penalty for a small gain; it is declining to flex at all. Flexibility in Gulf SWRO is not a marginal proposition that better control could tip, but a structurally dominated one. Second, even in the limiting case of completely harmless restarts ($N_{tol} \to \infty$), the benefit is only −0.002 USD/m³, about 0.2% of LCOW. Removing restart damage entirely does not unlock flexibility, because the part-load and stagnation terms of Eqs. (7)–(8) remain: reduced crossflow weakens shear removal, and idle hours foul faster still. The barrier is not one mechanism but three acting together. Third, the payoff is extraordinarily asymmetric. The maximum benefit of correctly-priced flexibility is −0.002 USD/m³; the maximum penalty for pricing it blind is 0.469 USD/m³, or 47% of LCOW, at $N_{tol} = 250$. The downside exceeds the upside by a factor of approximately 198. For an operator holding a 25 yr water purchase agreement this asymmetry is decisive independently of where the true N_{tol} lies: under uncertainty, steady operation is the rational choice.

TABLE III

Cycle-tolerance frontier. Δ is the degradation-aware flexible design relative to PV + steady baseload; negative means flexibility pays. The final row removes restart damage entirely. Tariff spread 0.25.

N_{tol}	yr@360	base	blind	aware	Δ
250	0.7	1.000	1.469	1.000	+0.000
500	1.4	1.000	1.228	1.000	+0.000
1000	2.8	1.000	1.109	1.000	+0.000
2000	5.6	1.000	1.049	1.000	+0.000
4000	11.1	1.000	1.019	1.000	+0.000
8000	22.2	1.000	1.004	0.998	−0.002
∞	—	—	—	—	−0.002

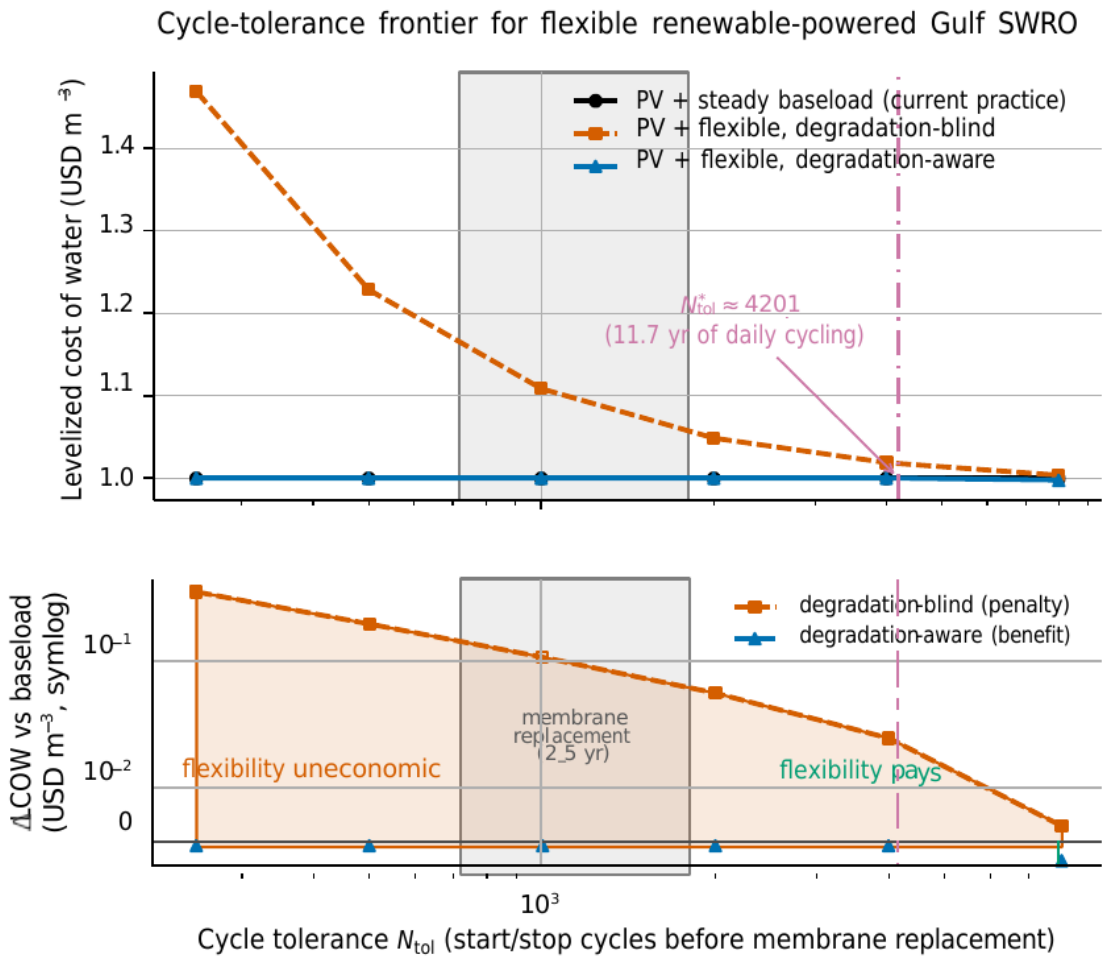


Fig. 3. Cycle-tolerance frontier. Upper panel: LCOW of the three strategies. Lower panel: ΔLCOW against steady baseload on a symmetric-log axis, showing the asymmetry between the benefit of correctly-priced flexibility and the penalty for pricing it blind. Shaded band marks the 2 yr to 5 yr membrane replacement interval reported for operating plants [40].

D. Robustness to biofouling severity

Sweeping the feed biofouling index B from 1.0 (open ocean) to 4.0 jointly with N_{tol} (Fig. 4) does not change the verdict. At every finite cycle tolerance tested the degradation-aware controller again selects zero flexibility for $B \geq 2$, and at $B = 1.0$ it does so for all $N_{tol} \leq 4000$. Only when restart damage is removed entirely does flexibility pay, and then by at most −0.021 USD/m³ at $B = 1.0$, falling to −0.005 USD/m³ at $B = 4.0$. The direction of that last trend is worth noting because it is counterintuitive: the benefit of flexibility shrinks as feedwater becomes more biologically active. The reason is that at high B the baseline plant is itself degrading quickly, so membrane replacement dominates the cost structure regardless of dispatch, and the marginal effect of how the plant is operated diminishes. Severe biofouling does not make flexibility attractive; it makes the whole question less relevant than fixing the feedwater.

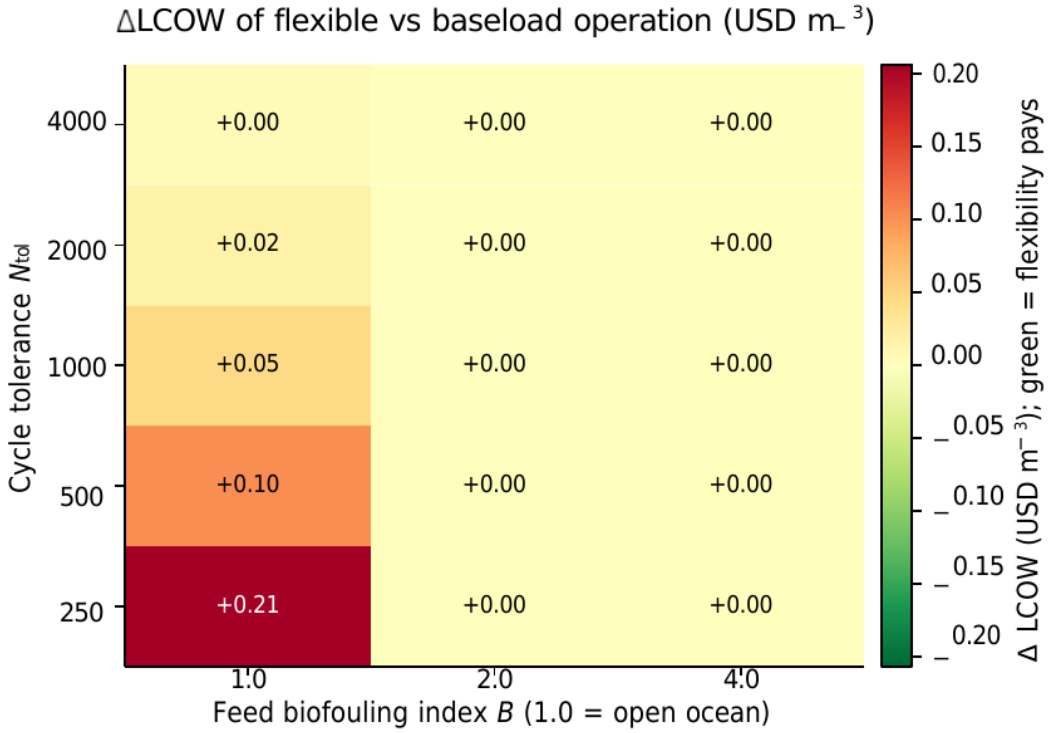


Fig. 4. ΔLCOW of degradation-aware flexible operation relative to steady baseload, across cycle tolerance and feed biofouling index. Green indicates flexibility pays.

E. Tariff spread does not rescue flexibility

A natural objection is that the Gulf's flat industrial tariff, rather than membrane degradation, is what suppresses flexibility, and that time-of-use pricing would change the answer. It does not (Fig. 5). Sweeping the tariff spread from 0 to 1.00 at fixed cycle tolerance, flexibility remains uneconomic at every finite N_{tol} tested. More strikingly, at low cycle tolerance a wider spread makes matters worse rather than better: at $N_{tol} = 500$ the penalty grows from 0.000 USD/m³ at zero spread to 0.165 USD/m³ at unit spread. The mechanism is straightforward. A larger spread increases the apparent arbitrage available to a flexible plant, drawing the controller into cycling that its membranes cannot support; the arbitrage is real but smaller than the degradation it provokes. Only when restart damage is removed entirely does tariff spread behave as the demand-response literature expects, improving the flexible case monotonically from −0.008 USD/m³ to −0.043 USD/m³. Tariff design is therefore not the lever that unlocks flexible desalination in

the Gulf. Introducing a strong time-of-use signal without first establishing membrane cycle tolerance would push operators toward a dispatch that destroys value.

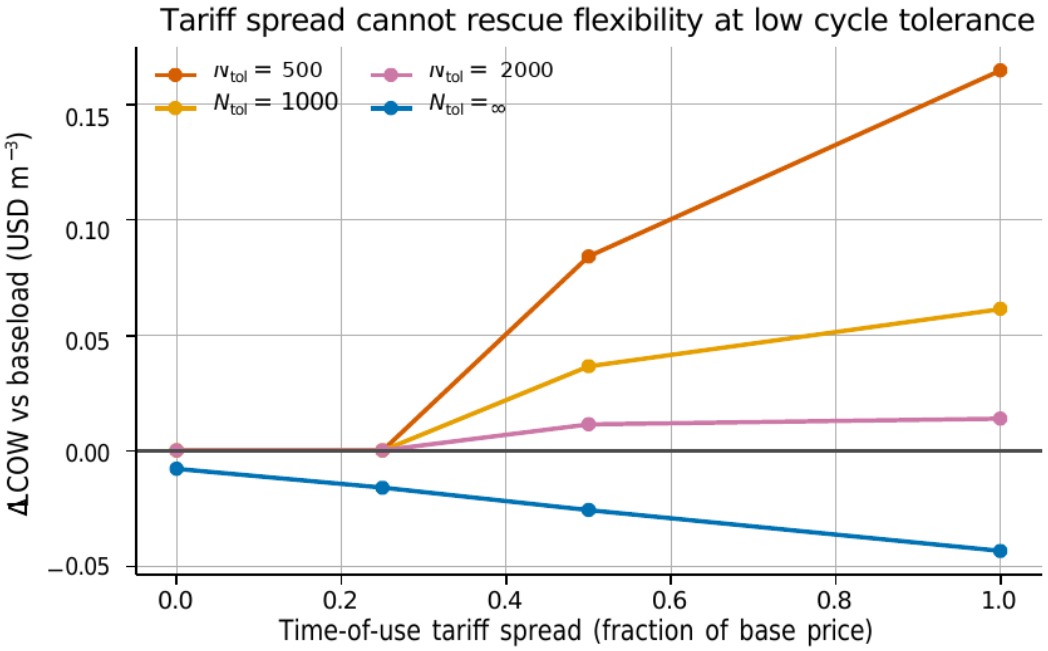


Fig. 5. Tariff-spread frontier. At finite cycle tolerance a wider time-of-use spread increases the penalty rather than reducing it, because it draws the controller into cycling the membranes cannot support.

TABLE IV

Renewable saturation and the implicit carbon price required to justify PV capacity, steady-baseload plant at over-size 1.09. Break-even prices are shown at the model's grid emission factor (EF) of 0.40 kg CO_2/kWh and at the UAE's 2019 baseline of 0.55 and 2030 target of 0.27 [41]; the price scales inversely with the factor.

PV (MWp)	RE (%)	LCOW (USD/m³)	Break-even EF 0.55 (USD/t CO_2)	Break-even EF 0.40 (USD/t CO_2)	Break-even EF 0.27 (USD/t CO_2)
0.0	0.0	0.732	—	—	—
10.5	30.2	0.750	28	38	57
21.0	39.8	0.783	61	84	124
35.0	43.4	0.835	112	155	229

F. Renewable saturation and the binding constraint

Sweeping PV capacity for a steady-baseload plant reveals a saturation that reframes the problem. Renewable fraction rises from 30.2% at 10.5 MW_p to 39.8% at 21.0 MW_p, but only to 43.4% at full energy parity (35.0 MW_p), while LCOW rises monotonically from 0.732 USD/m³ to 0.835 USD/m³ (Table IV). Doubling PV beyond 21.0 MW_p buys 3.6 percentage points of renewable fraction for 0.052 USD/m³. The mechanism is a curtailment wall. A plant running at steady load can absorb solar only during daylight; capacity beyond that is exported at half the import price. Passing the wall requires either storage or flexible operation, and Sections IV-B and IV-C show that flexibility is unavailable at any

realistic membrane cycle tolerance. It follows that the binding constraint on renewable penetration in Gulf desalination is membrane cycle tolerance, not the cost of solar generation and not the cost of storage. This is the paper's central claim, and it is a claim about membranes reached entirely through an energy-systems analysis.

G. The implicit carbon price of existing PV

No PV capacity is levelized-cost-optimal at UAE grid prices: the design-space optimum contains no PV at all (0.00× energy parity), at over-size 1.09 and LCOW 0.730 USD/m³, because grid electricity at 0.040 USD/kWh is cheaper than amortized photovoltaic generation at 650 USD/kWp. Existing deployments are therefore mandate-driven rather than cost-driven. Rather than treat this as an awkward result we invert it, computing the carbon price at which each PV increment breaks even (Table IV). At the model's grid emission factor the first 10.5 MW_p breaks even at 38 USD/t CO_2; across the range from the UAE's 2019 baseline to its 2030 target the same increment breaks even at 28 USD/t CO_2 to 57 USD/t CO_2, rising as the grid decarbonizes because each kilowatt-hour displaced then avoids less CO_2. Reaching 39.8% requires 84 USD/t CO_2, and 43.4% requires 155 USD/t CO_2. The steeply rising marginal abatement cost is the economic signature of the same curtailment wall: without flexibility or storage, each additional panel displaces less grid energy than the last.

H. Free design choice, and the self-assessment error

When both designers are allowed to choose the plant as well as its dispatch, they converge: each selects no PV and the minimum over-size, and the realized penalty for degradation-blindness collapses to 0.002 USD/m³. This is not evidence that degradation can be ignored. It is a consequence of PV being uneconomic at UAE grid prices in the first place: a designer free to build no solar never encounters the flexibility question.

The informative quantity in this case is instead the blind model's error about its own chosen design. Excluding membrane replacement from the objective, it reports an apparent LCOW of 0.651 USD/m³; the true cost of the same design is 0.733 USD/m³, an optimism of 12.6%. A model that omits degradation is therefore materially overoptimistic even about a plant it has correctly sized, and the error grows to 45% once a renewable mandate forces the flexible configuration of Table II.

I. Sensitivity

Fig. 6 reports one-at-a-time sensitivity of the baseline design over all 8 flagged parameters. The three largest are the feed biofouling index (LCOW range 0.385 USD/m³), the shear-removal exponent (0.114 USD/m³) and the PV capital cost (0.056 USD/m³).

Two parameters, the stagnation multiplier m_{idle} and the per-restart increment δR, have no effect on the baseline design. This is a correctness check rather than a null result: a steady baseload plant never idles and never restarts, so those terms cannot enter. It also isolates where they do matter. Both parameters act only once a plant flexes, which is precisely why a degradation-blind model calibrated on steady operation can appear perfectly well validated and still be badly wrong about load-following.

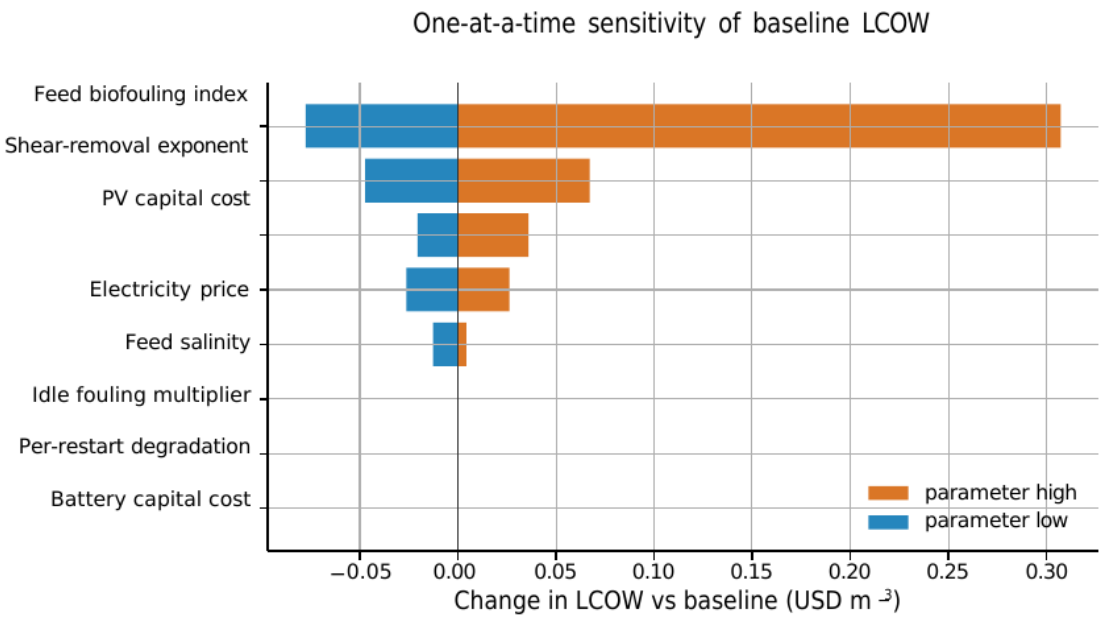


Fig. 6. One-at-a-time sensitivity of baseline LCOW. The stagnation multiplier and per-restart increment have no effect on a steady-baseload plant, and act only when the plant flexes.

V. Discussion

Operator behavior is rational, not conservative. The configuration our degradation-aware optimizer selects, namely co-located PV supplying roughly 39.5% of electricity with the RO trains held at steady load, is the configuration Gulf operators have actually built. Al Taweelah's solar park of about 69 MW_p, expected to supply at least 30% of the plant's electricity capacity and targeted to reach 55% [6], sits squarely in this regime, and its energy share of roughly 9% on our balance is the daylight-absorption limit of Section IV-F seen from the plant's side. An independent analysis of PV-following SWRO in Jordan reaches the same practical conclusion on capacity-utilization grounds alone [29]. The dispatch literature implicitly treats such plants as leaving value on the table by not load-following. Our results suggest the opposite: at every plausible cycle tolerance the flexible strategy is worse, and the operators' revealed preference encodes information about membrane degradation that the models omit. This is worth stating plainly because it inverts the usual direction of advice from modeling to industry.

The asymmetry is the actionable finding. We do not claim a point estimate of N_{tol}; none is available. What is robust is the shape of the payoff: at most 0.2% upside against up to 47% downside, a ratio near 198 to one. Decision-making under that asymmetry does not require knowing N_{tol} precisely; it requires only knowing that it might be low. Any operator facing a long-duration water purchase agreement should therefore default to steady operation until the cycling penalty is measured for their feedwater. Conversely, a developer who has measured a high N_{tol} for a specific membrane and feed has a defensible basis for flexible operation, and Fig. 3 states what it is worth.

Where the remaining headroom actually is. Our results locate the opportunity away from where the framing of "low-energy desalination" usually points. Process-level specific energy is near its floor and part-load operation makes it worse, not better. The 0.452 USD/m³ of avoidable membrane cost in Table II is an order of magnitude larger than the 0.021 USD/m³ of accessible energy saving. Effort directed at cycling-tolerant membranes, or at restart protocols that suppress the irreversible increment δR (permeate flushing before shutdown is the obvious candidate), would unlock more value than further pursuit of

specific energy consumption. The same conclusion follows from the $N_{tol} \to \infty$ row of Table III: even eliminating restart damage entirely is not sufficient, so part-load and stagnation behavior must be addressed alongside it.

Policy implications. Four follow directly. First, because PV at this site is mandate-driven rather than cost-driven, the instrument that matters is the carbon price or equivalent obligation, and Table IV gives the schedule: 38 USD/t CO_2 buys 30.2% renewable fraction, 155 USD/t CO_2 buys 43.4%. Second, the steeply rising marginal abatement cost means that mandating renewable fraction above roughly 39.8% without addressing the curtailment wall procures emissions reductions at rapidly escalating cost. Third, as a caution rather than a recommendation, introducing strong time-of-use price signals for desalination in the Gulf would, on our results, push operators toward value-destroying dispatch unless membrane cycle tolerance is established first. Fourth, and most usefully, the highest-leverage public investment identified by this study is not generation or storage capacity but a membrane cycling-tolerance testing program, which is inexpensive relative to the capital decisions it would inform.

VI. Limitations

We state these directly because the conclusions should be read through them.

This is a framework, not a validated plant model. The model is benchmarked against published values rather than operating records: its process energy use lies between the lowest plant value on record [37] and a full-scale Gulf projection [4], its feed pressure and energy-recovery efficiency are consistent with reported ranges [34], and its osmotic pressure and cleaning interval are consistency checks rather than calibrations (Table I). Gulf-condition outputs, including the predicted 1.6 months cleaning interval at B = 2 and 30 °C, are model predictions we cannot presently test.

Most parameters are assumptions. Of 52 model parameters, 34 (65%) are assumptions of this study, 9 are standard engineering values that we have not individually sourced, and only 5 are taken directly from verified literature; the rest are derived or follow industry practice. Appendix A tags every one. A verification pass against publisher records moved several parameters from “literature” to “assumption” or “engineering value” because their original sources could not be confirmed. This is why results are reported as functions of the uncertain parameters and why the central claim is expressed in units (N_{tol}) chosen to be refutable.

The decisive parameter is unmeasured. We could locate no measurement of per-cycle irreversible permeability loss for warm hypersaline seawater. The closest evidence, from short laboratory campaigns on brackish and seawater membranes [16], [18] and a 14-year intermittent brackish plant without a continuous control [17], establishes that a penalty exists but does not separate irreversible from recoverable loss per cycle. Conversely, the only seawater campaign with a comparable number of cycles that we located found no significant membrane damage after 1,568 one-hour start–stop cycles [21], which suggests that, at least for short stops, the cycle tolerance exceeds the number of cycles tested. Our default δR corresponds to $N_{tol} = 264$, which is an assumption, not a finding. This default is pessimistic relative to that campaign, which is why the conclusions are reported across the frontier of Table III rather than at the default value. The frontier is reported precisely so a measured value can be substituted.

The fouling model is phenomenological. Equations (7)–(8) impose plausible functional forms (a Q_{10} law, a power-law shear dependence, and a constant stagnation multiplier) on mechanisms that are in reality coupled and spatially distributed. We do not resolve biofilm structure, and the shear exponent α and stagnation multiplier m_{idle} are not independently identified. Because the $N_{tol} \to \infty$ result shows these two terms alone are sufficient to suppress flexibility, their joint identification is as important as measuring δR.

Discretization of the policy search. The degradation-aware designer searches a coarse grid of flexibility aggression. Where the frontier reports $\Delta = 0$ exactly, the optimizer has selected the degenerate zero-flexibility member of that family; a finer grid could in principle find a small positive benefit that we would not resolve. This biases against flexibility only in the third decimal place, which is immaterial against the penalties reported here, but it should be stated.

Scope restrictions. One resource year (2023), one site, one representative train, one membrane technology, single-stage continuous SWRO with isobaric energy recovery. Brine disposal is not costed, which matters in a semi-enclosed basin where brine raises ambient salinity [22]. Battery storage is implemented but not optimized jointly with the dispatch policy. Membrane replacement is triggered at a normalized permeability of 0.70, whereas a full-scale SWRO plant has operated for seven years with a 40% decline in normalized flux [35]; real replacement practice may tolerate larger losses, which would lower replacement cost but raise the energy penalty of fouled operation. The grid emission factor of 0.40 kg CO_2/kWh is an intermediate value between the UAE's 2019 baseline and 2030 target [41]; Table IV reports the break-even carbon price across that range. The intake temperature cycle of 20 °C to 35 °C is narrower than the reported surface range of below 15 °C to 36 °C [22], and the salinity sweep stops below the 47.5 g/L to 53.5 g/L recorded at a Gulf pilot intake [4]. Colder winters would raise winter specific energy and slow winter fouling, and higher salinity raises energy use throughout; feed salinity is among the least influential parameters in Fig. 6, and load-following decisions arise in sunlit periods, so neither is expected to reverse the conclusion, but neither has been tested directly. The base-case recovery of 40% is above the 31–34% reported for stable operation on high-salinity, high-temperature seawater [42]; on the paper's own Eqs. (1) and (5), process specific energy changes by at most 0.03 kWh/m³ over that range.

Capacity share versus energy share. Published descriptions of Al Taweelah's solar contribution are stated as a share of electricity capacity [6]. Our energy-basis estimate of about 9% is not a published figure: it depends on our assumed site yield and specific energy, and we use the plant only as a design reference point.

VII. Conclusions and Further Work

We coupled a Gulf-parameterized biofouling and irreversible-degradation model to an hourly renewable dispatch and storage co-optimization for seawater reverse osmosis, addressing a gap that the current state of the art in desalination dispatch identifies in its own future work. Five findings stand out. First, the Arabian Gulf presents an opposition between energy cost and fouling rate: specific energy consumption is minimized in summer because permeability rises with temperature, exactly when biofouling is fastest. Second, degradation-blind load-following, the behavior assumed by existing dispatch models, raises renewable fraction from 39.5% to 52.3% while raising levelized water cost by 45%, because membrane life falls from 2.06 yr to 0.53 yr; a degradation-aware controller with identical structure and foresight instead selects zero flexibility. Third, this holds across the entire plausible range of membrane cycle tolerance: the critical value is $N_{tol}^{*} \approx 4201$ cycles, or 11.7 yr of daily cycling, more than twice the 2 yr to 5 yr after which operating plants replace their membranes, and even removing restart damage entirely yields at most 0.2% benefit because part-load and stagnation penalties remain. Fourth, the payoff is asymmetric by a factor near 198, which makes steady operation the rational default under uncertainty regardless of the true tolerance. Fifth, renewable fraction saturates near 43.4% for a steady-baseload plant even at full PV energy parity, so the binding constraint on renewable penetration in Gulf desalination is membrane cycle tolerance rather than the cost of solar generation or storage.

The experiment that would settle this. Our conclusion is designed to be refutable by a single measurement. We propose: take a standard seawater RO element on Gulf feedwater (45 g/L, 30 °C to 35 °C, with representative biological load); operate it under two regimes in parallel, one continuous and one with a daily start/stop cycle including a realistic shutdown duration; normalize permeability for

temperature and pressure; apply identical cleaning protocols to both; and record the permeability not recovered by cleaning as a function of cumulative cycle count. The slope of that curve is $\delta R\, f_{irr}^{start}$, and Eq. (11) converts it to N_{tol}. A campaign of a few hundred cycles (under a year at daily cycling, and less under accelerated protocols) would place a real plant on Fig. 3. The same rig, run at reduced load rather than stopped, would identify the shear exponent α, and held idle would identify m_{idle}; our results show all three are needed, because eliminating restart damage alone is not sufficient.

Other further work. Joint optimization of battery and product-water storage with the dispatch policy; restart protocols, particularly permeate flushing, treated as a decision variable rather than a fixed practice; extension to the legacy thermal desalination fleet, whose displacement is a larger emissions lever than marginal renewable integration; and inclusion of brine disposal cost, which in a semi-enclosed basin is a genuine externality that our LCOW omits.

DECISION-SUPPORT DASHBOARD

The results are also delivered as an interactive decision-support dashboard (Fig. 7), a single self-contained web page built on the same result files as this paper. A viewer sets the membrane cycle tolerance, feed biofouling index, time-of-use tariff spread, and feed conditions. The verdict, the cycle-tolerance frontier, the comparison of the three operating strategies, and a break-even gauge are looked up from the hourly simulations and interpolated between computed points, while specific energy, feed pressure, and the time to the first cleaning are recomputed live from the model's closed-form relations. The dashboard adds no new results; it lets a jury member, operator, or regulator explore the assumptions of Table V without rerunning the model.

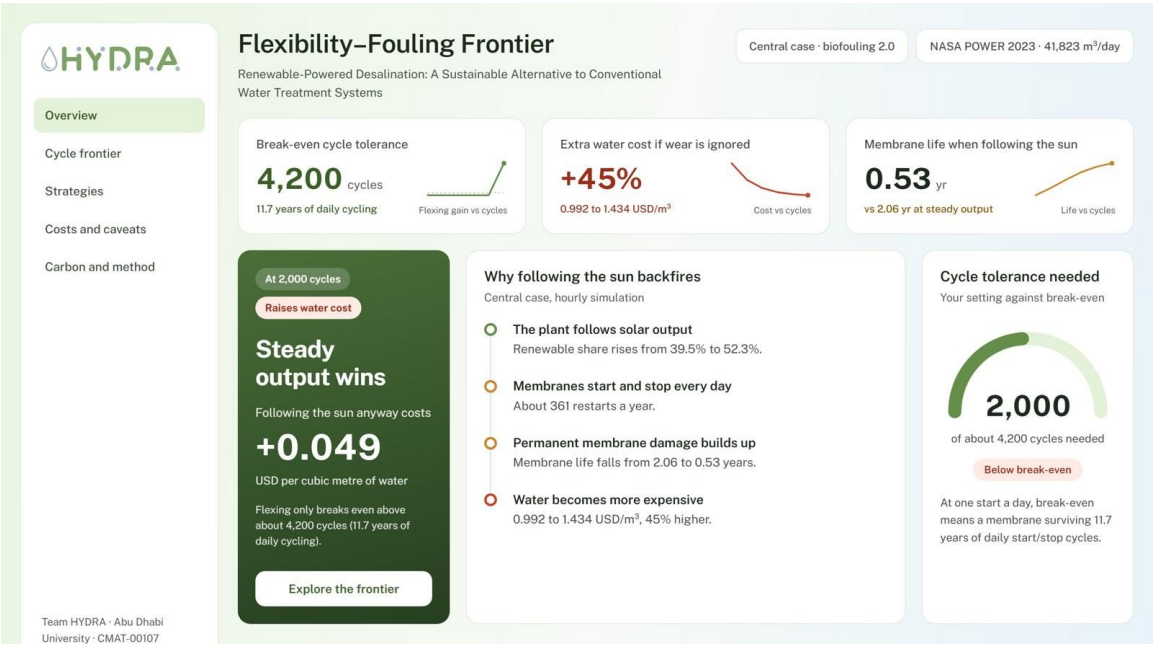


Fig. 7. Overview screen of the HYDRA decision-support dashboard at the central case (biofouling index 2.0, cycle tolerance 2,000). The headline figures, verdict, cause chain, and break-even gauge are drawn from the same simulation results as Fig. 3.

REPRODUCIBILITY

All code, parameters, and results are available from the corresponding author on request. The resource data are public [33]. python physics.py and python fouling.py print the benchmark and consistency checks of Table I; python run_experiments.py re-generates every number reported here into results/*.csv;

python make_figures.py re-generates every figure from those files without re-simulation. The parameter appendix is generated from the configuration file so it cannot drift from the values used.

APPENDIX A: MODEL PARAMETERS AND PROVENANCE

TABLE V
Model parameters and their provenance.

Group	Parameter	Value	Provenance
Gulf seawater	Feed salinity S_f (g/L)	45	ASM
Gulf seawater	Min sea-surface temperature (°C)	20	ASM
Gulf seawater	Max sea-surface temperature (°C)	35	ASM
Gulf seawater	Osmotic coefficient φ	0.93	ENG
Gulf seawater	Feed biofouling index B	2	ASM
RO process	Recovery ratio r	0.4	ASM
RO process	Clean permeability A25 (L m^{-2} h^{-1} bar^{-1})	1	ENG
RO process	TCF constant Ke (K)	2700	IND
RO process	Design flux J (L m^{-2} h^{-1})	13	ENG
RO process	Salt rejection R	0.995	DER
RO process	HP pump efficiency ηHPP	0.85	ENG
RO process	ERD efficiency ηERD	0.94	LIT
RO process	Feed-channel ΔP (bar)	2	ASM
RO process	Auxiliary SEC (kWh m^{-3})	0.5	ENG
RO process	Minimum stable load Lmin	0.4	ASM
Fouling and degradation	Biofouling rate coefficient kbio	0.045	ASM
Fouling and degradation	Growth temperature sensitivity Q10	2	ENG
Fouling and degradation	Shear-removal exponent α	0.8	ASM
Fouling and degradation	Stagnation multiplier midle	2.5	ASM

Group	Parameter	Value	Provenance
Fouling and degradation	Per-restart increment δR	0.004	ASM
Fouling and degradation	Irreversible fraction f_{irr}	0.12	ASM
Fouling and degradation	Irreversible fraction, restart f_{start}	0.35	ASM
Fouling and degradation	Scaling coefficient k_{scale}	0.01	ASM
Fouling and degradation	CIP trigger (normalized permeability)	0.85	ENG
Fouling and degradation	CIP recovery fraction	0.92	ASM
Fouling and degradation	CIP cost (USD)	4500	ASM
Fouling and degradation	Minimum CIP interval (h)	720	ASM
Fouling and degradation	Replacement threshold prep	0.7	ASM
Fouling and degradation	Membrane life ceiling (yr)	5	ENG
Photovoltaics	Module efficiency at STC	0.205	ASM
Photovoltaics	Power temperature coefficient (K^{-1})	−0.0034	ENG
Photovoltaics	NOCT (°C)	45	ASM
Photovoltaics	DC/AC ratio	1.2	ASM
Photovoltaics	Soiling rate per unit AOD per day	0.0045	ASM
Photovoltaics	Soiling loss ceiling ℓ_{max}	0.25	ASM
Photovoltaics	PV wash interval (d)	14	ASM
Photovoltaics	PV wash cost (USD m^{-2})	0.35	LIT
Storage	Battery round-trip efficiency	0.9	ASM
Economics	Real discount rate i	0.055	LIT

Group	Parameter	Value	Provenance
Economics	Economic life n (yr)	25	LIT
Economics	Grid electricity (USD kWh^{-1})	0.04	LIT
Economics	PV export value (USD kWh^{-1})	0.02	ASM
Economics	Time-of-use spread	0	ASM
Economics	Grid emission factor (kg kWh^{-1})	0.4	DER
Economics	RO CAPEX (USD per $m^3\ d^{-1}$)	1200	ASM
Economics	PV CAPEX (USD kW^{-1})	650	ASM
Economics	Battery CAPEX (USD kWh^{-1})	220	ASM
Economics	Tank CAPEX (USD m^{-3})	90	ASM
Economics	Membrane cost (USD m^{-2})	22	ASM
Economics	Fixed O&M (USD m^{-3})	0.22	DER
Water demand	Mean utilization of nameplate	0.92	ASM
Water demand	Seasonal demand amplitude	0.12	ASM

LIT published literature, IND industry disclosure, DER derived or calibrated, ASM assumption of this study, ENG standard engineering value not individually sourced, DATA measured reanalysis.

Every ASM entry is swept in the sensitivity analysis; the three governing the flexibility conclusion are α, m_{idle}, and δR.

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